

Analytathon 1

Quintin Quinn

1 INTRODUCTION

The energy market in Ireland and Northern Ireland is a single energy market (ISEM) that is used to help increase competition across energy companies, but most importantly, supply energy to all homes and businesses within the island at a fair price. Generators bid to supply power and are dispatched based on the lowest cost to meet real-time demand. Prices are set marginally, meaning the most expensive unit needed to meet demand determines the market price, which is then paid to all dispatched generators. It is also highly interconnected with the UK energy market. The ISEM contains the following 5 markets:

- **DAM:** The Day-Ahead Market(DAM) allows suppliers to buy and sell electricity one day before delivery, with prices set for every 30-minute interval. These prices are primarily driven by forecasted demand and expected renewable generation, particularly wind. Generators submit bids, and the market clears based on the lowest-cost combination needed to meet the predicted demand for each period.
- **IDA1:** This is the first Intraday Auction(IDA1), opening shortly after the DAM gate closes at 11:00 AM on the day before delivery. It provides participants with another opportunity to adjust their positions closer to real-time, still within the day-ahead window. All bids must be submitted by 5:30 PM
- **IDA2:** This auction works the same way as IDA1, but it runs from 11:00 to 23:00 on the day of delivery. All bids must be submitted by 08:00 that morning.
- **IDA3:** Again, this operates just like IDA1 and IDA2, but it runs from 17:00 to 23:00. All bids must be submitted by 14:00.
- **Balance Market:** The balancing market(BM) is a real-time electricity market, and it's also the most volatile.

1.1 OUR OBJECTIVE

For this Analytathon, Energia Group tasked us with developing a machine-learning algorithmic trading strategy for the ISEM. They provided a comprehensive dataset covering the years 2022–2025, containing market prices from all trading windows mentioned above, alongside a wide range of supporting variables such as predicted demand, weather-related forecasts, interconnector flows between Ireland and Great Britain, and detailed information on wind and solar generation.

Our goal was to work through this high-dimensional dataset, isolate the features that genuinely contribute to profitable decision-making, and build a strategy that maximises average euro per megawatt hour (€/MWh). In addition to improving profitability, we also needed to understand and mitigate risk by identifying the conditions under which the model performs well, as well as the scenarios where losses are more likely to occur. Also, we need to make sure our total profit and loss value is maximised.

2 SETTING UP THE MODEL

2.1 INITIAL STEPS

Before modelling, we first prepared the dataset by loading, cleaning, and standardising it for analysis in Python. After converting the original Parquet file to CSV, we ordered the data chronologically, normalised the datetime format, and removed invalid or duplicate entries. A major issue was the large number of missing values, more than 26% of the dataset, which came mainly from IDA2 and IDA3 since these markets only operate during limited hours. Because these gaps were structural rather than random, we removed IDA2 and IDA3 entirely. This allowed us to work with a consistent and complete dataset and focus our modelling on markets with full daily coverage: the DAM, IDA1, and the BM.

2.2 MARKETS AND CORRESPONDING VARIABLES

Since our model predicts IDA1 or BM prices using DAM information, we restricted the feature set to variables available day ahead. This meant removing any real-time data such as actual demand or generator output, ensuring there was no information leakage and the model reflected realistic forecasting conditions and reducing missing values. After this filtering, 24,804 missing entries remained, almost all from the variable 'ISEMWIND_U_50_DAM', which

contained too many gaps to use reliably and was therefore removed. With a clean dataset established, we examined which remaining features showed meaningful relationships with our target markets. A correlation heat map, shown in Figure 1, was generated using all day-ahead accessible variables, and we selected those with the strongest and most relevant correlations to DAM and BM prices. These variables, included in the heat-map below capture a mix of key market drivers, including electricity price signals from Great Britain's energy markets, wind and solar generation forecasts, system-wide demand forecasts, and probabilistic wind uncertainty measures. Together, they will help train our models.

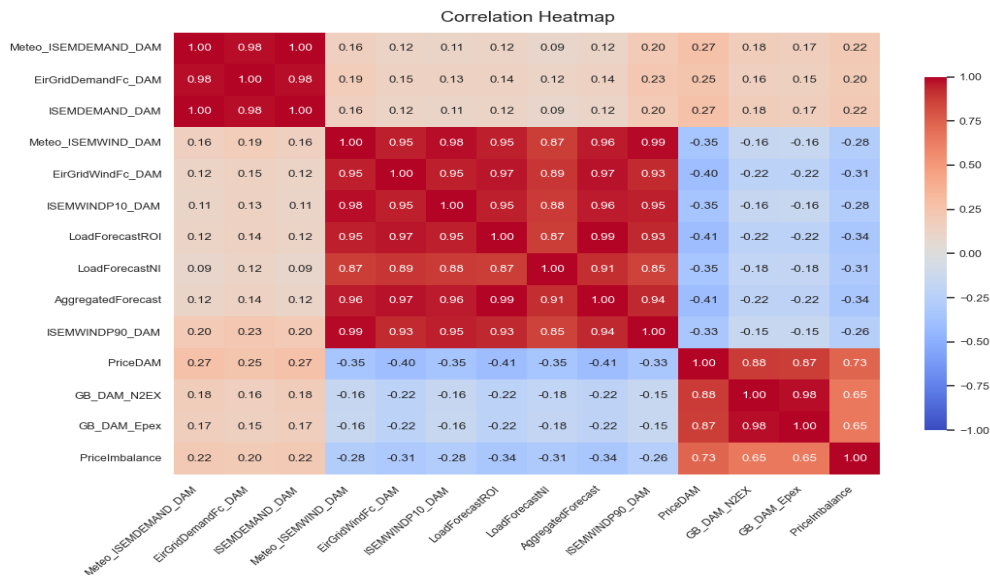


Figure 1: Correlation heat-map of all day-ahead accessible variables.

2.3 IMPORTANT DEFINITIONS

Our final steps before training the model included creating some important definitions. We created a profit variable as well as a profit-direction variable so the model could learn the outcome it was trying to predict. We defined the spread as the difference between the DAM and either the IDA1 or the BM market prices. As for our profit-direction variable, we assigned +1 when the later price was higher and -1 when it was lower. This produced our classification target, y_dir, allowing the model to focus on predicting price direction rather than magnitude. We also engineered two additional features to give the model a clearer sense of system conditions: one representing demand remaining after renewable generation, and another capturing uncertainty in the wind forecast. These definitions are used later in the modelling process. Our final key definition was defining the training and test sets. We found that models performed best when the test set consisted of one continuous year, ensuring realistic temporal separation. The training period therefore, covered 01/01/2022–31/12/2024, while the test set spanned 01/01/2025–31/12/2025, providing a clean and chronologically consistent split.

3 MODEL SELECTION

The next stage of our process was deciding on what models may help us yield the best results. After some research, along with our prior knowledge in some past modules in machine learning, the 3 models we decided on were:

3.1 RANDOM FOREST

Random Forest builds many decision trees, each trained on different random subsets of data and features. At each split, only a small random selection of variables is considered, which reduces overfitting and increases model robustness. The final prediction is made by aggregating all trees, giving strong performance on complex, non-linear datasets. [1]

3.2 EXTREME GRADIENT BOOSTING

This model, much like Random Forest, also relies on decision trees, but instead of building them independently, it constructs them sequentially. We also draw on the concept of gradient boosting, where many weak models are combined in a way that each new tree focuses on correcting the errors of the previous ones. The “gradient” aspect comes from using the gradient of the loss function to guide how each new tree should adjust its predictions, allowing the model to steadily improve and reduce errors over time. [2]

3.3 LOGISTIC REGRESSION

Logistic Regression estimates the probability that an input belongs to a particular class by applying the sigmoid function to a linear combination of the input features. The sigmoid maps any real value into the range 0–1, making it ideal for probability-based classification. The model outputs the probability of the positive class, with the complementary probability representing the negative class. It's important to remember when implementing these models that we scale all data. Proper scaling helps Logistic Regression converge reliably and prevents features with larger magnitudes from dominating the model. [3]

4 RESULTS

4.1 BASIC RESULTS

We began by testing the basic model. With no restrictions or any alterations to the model. As mentioned before, for these basic models, we will test them on both DAM to IDA1 as well as trades from 'DAM' to 'BM' to see what yields the best results. Firstly, for DAM to IDA1:

Model	Accuracy	No. Trades	Profit (€)	€/MWh
Random Forest	0.549	17516	13851.38	0.791
XGBoost	0.551	17516	33041.95	1.886
Logistic Regression	0.545	17516	41472.3	2.368

Now for the trades between the DAM and BM:

Model	Accuracy	No. Trades	Profit (€)	€/MWh
Random Forest	0.555	17516	63245.14	3.611
XGBoost	0.557	17516	96898	5.532
Logistic Regression	0.549	17516	106471.54	6.079

All models executed the exact same number of trades because we did not apply any restrictions on trade selection. Every model was allowed to take every available trading opportunity. In total there were 17536 possible time steps, but twenty of these were missing essential feature values and had to be removed during preprocessing. This left 17516 valid timestamps. Trading across both market pairs shows very similar accuracy, but profitability tells a different story. The total profit and €/MWh are consistently higher when trading from the DAM to the BM, which indicates that this spread offers more valuable opportunities than DAM to IDA1. What initially seemed confusing was that accuracy did not align with profit. The highest earning model in the DAM to BM case was logistic regression, even though it was the least accurate of the six models, although only slightly. After reading [4], it became clear that this behaviour is common and that highly accurate models do not always generate higher returns, a phenomenon Mostafa Shams refers to as the Accuracy Paradox. From these results, it became clear that trading from the DAM to the BM is more lucrative over the year. This is likely due to the greater price volatility between these two markets since the longer time gap gives conditions more time to shift and creates larger and more frequent price spreads than those seen between DAM and IDA1. To check whether this pattern was consistent throughout the year, we examined cumulative plots of the three models for each market pair. From all three graphs, it is evident that the models trained and tested on

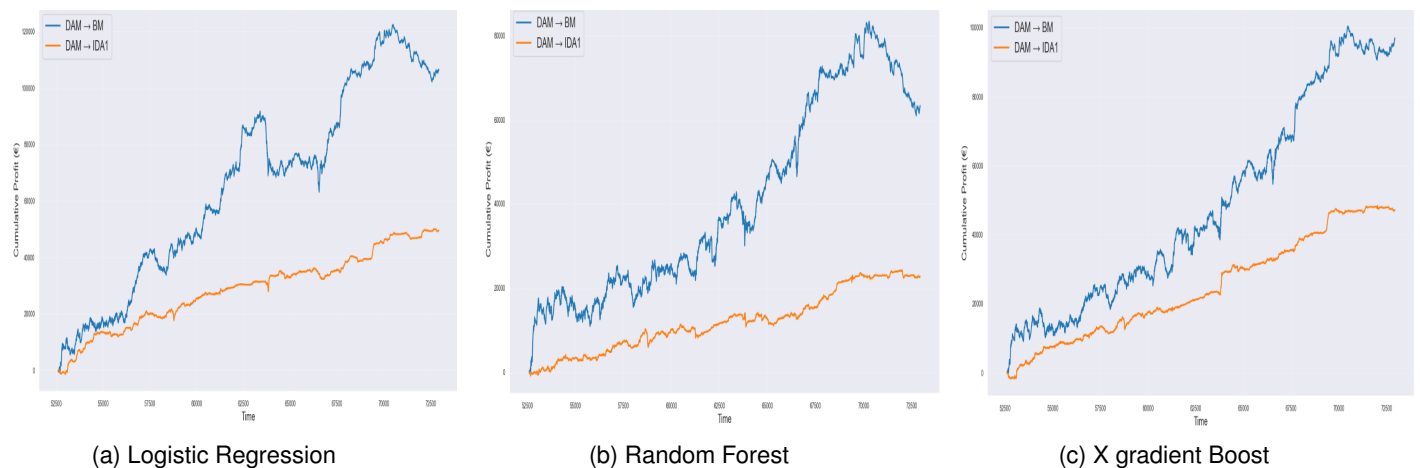


Figure 2:
Cumulative P&L for all models: DAM→BM (blue) vs. DAM→IDA1 (orange)

the DAM → BM trading route (blue) consistently yielded higher profitability across the entire timeline. This suggests

that both total profit and €/MWh are maximised when trading between the DAM and BM markets. The results from the DAM → BM route were compelling enough to lead us to focus all subsequent analysis exclusively on this trading route.

4.2 IMPROVING OUR RESULTS

The next step was to determine how to improve the model's performance. We considered reducing the number of trades the model executes, as taking more trades inherently increases the risk of making poor decisions. Fewer trades would theoretically reduce this risk. However, simply lowering the number of trades could lead to missing out on some of the best opportunities. This is when the idea of threshold restriction became a clear solution. By ensuring the model only selects trades it is most confident in, those with the highest probability of success according to our trained model, we could minimise risk while also improving our €/MWh.

To determine how strict this confidence filter should be, we plotted both cumulative profit and €/MWh across a range of trade-rate thresholds. Figure 3 shows these results for all three models, with the blue axis representing €/MWh and the orange axis representing total profit.

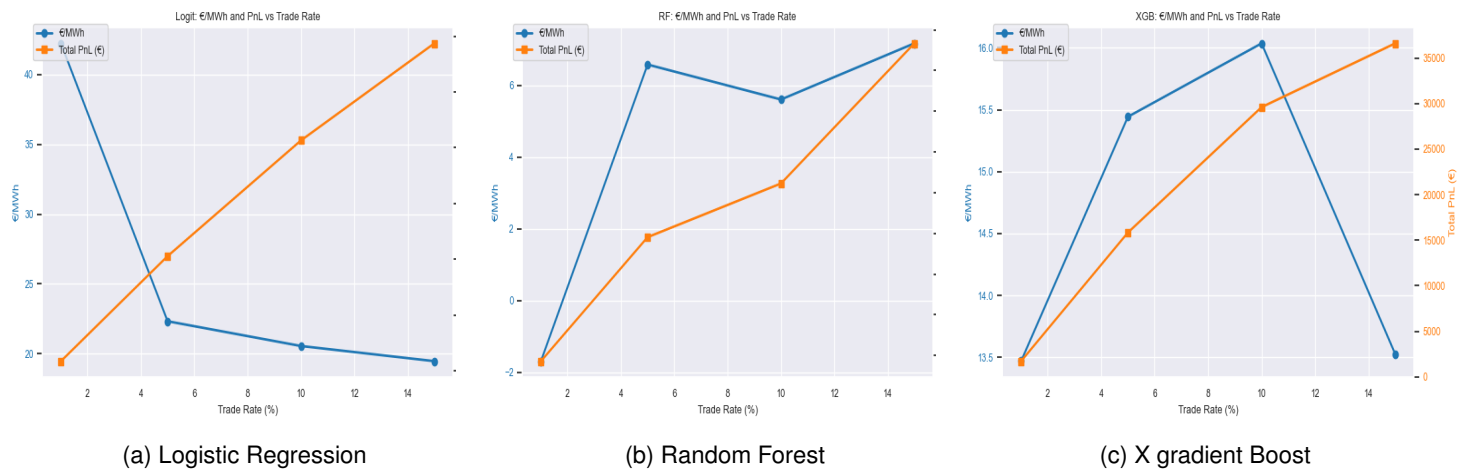


Figure 3: P&L in Blue and €/MWh in orange

Similar to the findings in Section 4.1, the logistic model again delivers the most balanced performance across nearly all levels of trade selectivity. Figure 3a shows a clear pattern: as we relax the selectivity requirement and allow the model to take more trades, the total profit increases, but €/MWh steadily declines. This behaviour is intuitive. When the model is allowed to trade more frequently, it captures more profitable opportunities overall, but it also includes a greater number of lower-quality trades. There are clear percentages that return the best results for our random forest and XgBoost models (15% and 10% respectively), but due to their €/MWh value always being lower than that of the Logistic regression, we had decided that it was our best performing model and it would be the first graph we use to decide our optimal threshold percentage.

4.2.1 RESULTS

The intersection point is the most sensible choice for setting our threshold percentage for the logistic regression model, so that is the value we use. At this point, the total profit is still rising strongly while the drop in profit per trade has already begun to level off, which makes it the most balanced option. The chosen percentage is 4.2%. The results are as follows,

Model	Accuracy	No. Trades	Profit (€)	€/MWh
Logistic regression	0.5488	2047	41758.1	20.4

Comparing these results to the results from section 4.1, the number of trades the model now takes has decreased by 15469, the total profit has decreased by €64,712,84 solely down to the amount of trades we are now taking, and our profit per trade has increased massively from €6.079 to €20.4.

4.3 UNCERTAIN WIND CONDITIONS

The Irish energy market is heavily influenced by weather forecasts. With over 30% of Ireland's electricity coming from wind generation, traders rely on accurate wind predictions to make informed decisions. However, Ireland's position as a small island on the edge of the Atlantic makes wind conditions particularly difficult to forecast. We suspected that this uncertainty might create trading opportunities. To investigate this, we examined whether higher profits tend

to occur when the weather is easy to predict or when conditions are more volatile. As mentioned in section 2.3, we created a variable called 'wind_uncertainty', it is defined as:

$$isem_wind_p90_dam - isem_wind_p10_dam$$

The P90 forecast represents a level of wind generation that actual output is 90% likely to stay below, while the P10 forecast represents a level that actual output is only 10% likely to fall below. By subtracting these two values, we are essentially finding out how confident the forecast is in predicting the correct wind speeds.

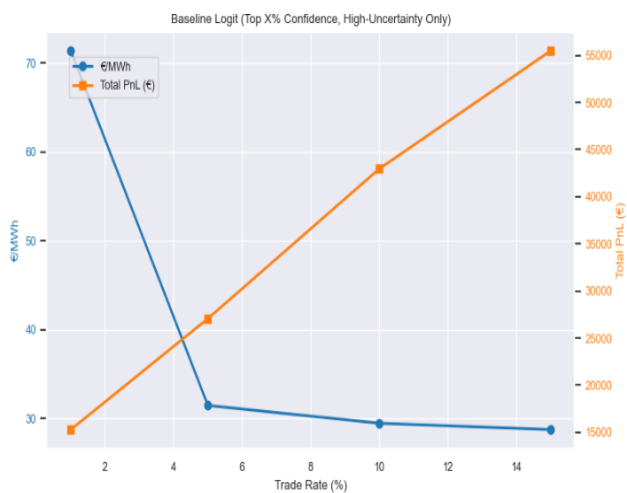
We divided wind-forecast uncertainty into four quartiles. Low, Med-Low, Med-High, and High, we used our model from Section 4.2 to analyse performance across these regimes.

Wind Uncertainty	€/MWh	No. Trades
Low	10.682500	52
Medium-Low	26.695700	400
Medium-High	17.590647	1005
High	33.440663	2747

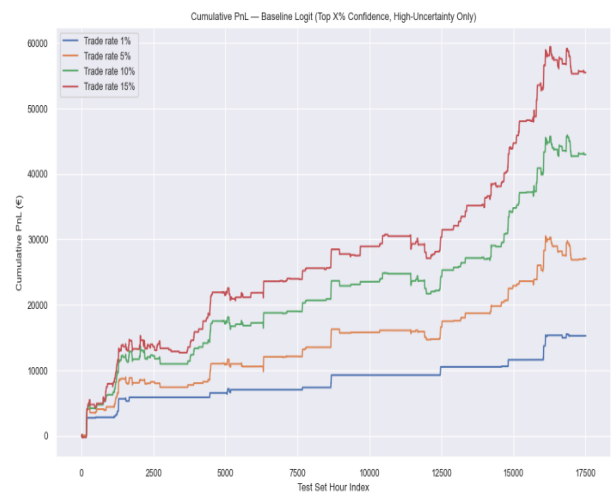
Our results showed that both trade frequency and profit per trade were highest during periods of high wind uncertainty. This naturally led us to focus on strategies that trade more aggressively in these conditions. The model we found most effective is a slight variation of the one in Section 5.2.1: after setting a confidence threshold, we execute only those trades that occur during high-uncertainty hours. Because these periods are more volatile and carry greater risk, the confidence filter remains essential—we trade only when the model is at its most certain. To identify the optimal threshold for maximising €/MWh, we repeated our earlier procedure and plotted €/MWh and total profit across different confidence levels. Figure 4a shows the resulting trade-off, and Figure 4b displays the corresponding cumulative profit.

Looking at Figure 4b below, there is a steady rise through most of the year, followed by a sharp increase and then a sudden decline near the end. These major declines represent the maximum drawdowns, which measure the loss from the highest peak to the lowest trough. In the one percent case, there are very few, if any, drawdowns at all, and the curve moves upward almost continuously. The four maximum drawdowns were as follows:

	1%	5%	10%	15%
Max Drawdown(€)	570.67	3620.35	3265.48	4104.14



(a)



(b)

Figure 4: Total P&L in Blue and €/MWh in orange

There is a point in every plot in Figure 4b where the rate of profit slows noticeably in the middle of the year. To understand this, I examined the percentage of trades occurring throughout the year, and the results below helped clarify this behaviour.

The values within this chart represent the number of trades that take place within each quarter at the 4 different thresholds.

Threshold (%) \ annual quarters	Q1	Q2	Q3	Q4
1	72	72	15	55
5	259	190	84	327
10	469	281	176	532
15	644	356	232	698

In every case, most trades occurred in the first and final quarters of the testing period. This matches the idea that the testing set spans a full annual cycle and that the opening and closing months in Ireland tend to have the most uncertain wind conditions. These conditions create more opportunities for the model to trade, and they are also the periods where the cumulative sum plots show the most activity. Since our main metric to optimise was €/MWh, our final results were based on the 1% threshold results you see below:

Model	Accuracy	No. Trades	Profit (€)	€/MWh
Logistic regression	0.7897	214	15272.29	71.37

It of course makes sense that there are less trades (214) as we are only taking the top 1% of all possible trades, but since we are solely trying to maximise €/MWh, this is a correct approach.

5 CONCLUSION

In conclusion, we found that the day ahead of balancing market pathway offered the strongest opportunity for profit within a single market trading system. This reflects both the large volume of available data and the full day between the two markets, which increases the chance of meaningful price differences. We also observed that logistic regression consistently outperformed the random forest and XGBoost models. This suggests that the relationship between DAM and BM prices is largely linear and that the simpler model generalises more effectively than the more complex nonlinear alternatives. Finally, we learnt that focusing on the periods when the most money is moving and restricting the model to only its highest quality trades leads to stronger overall performance.

REFERENCES

- [1] DSA 8003, "Introduction to Regression Models," https://canvas.qub.ac.uk/courses/33616/files/6705300?module_item_id=1579971.
- [2] Nvidia, "What is XGBoost?" <https://www.nvidia.com/en-gb/glossary/xgboost/>.
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- [4] Mostafa Shams, "A Statistical Evaluation of Machine Learning Profitability in the English Premier League," <https://www.qeios.com/read/7A61RS>.